

Acknowledgments

I dedicate this thesis to my parents, Patrizia and Livio, whose love and belief in me have helped shape who I am today. I am deeply grateful to my siblings, Alessandro and Sara, and to the rest of my family, for their constant support and care. I would like to extend my deepest gratitude to Prof. Colla, whose guidance and support were crucial for writing this thesis. Lastly, I thank my best friend Alessandro, whose friendship I deeply cherish and who has made this journey more memorable.

"As long as I live, there are infinite chances."

– A quote that has always inspired me

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1 Introduction

Asset pricing has long been one of the central themes in financial economics. Over the years, researches have proposed a variety of models that attempt to explain the cross-section of stock returns, with new variations continuously being developed and debated. Among the most influential contributions there are the three-factor model proposed by Fama and French (1993) and its later extension, the five-factor model (Fama and French, 2015). Both these models have become a benchmark in asset pricing and remain widely used in both academia and practice. Since their introduction, a large body of research has emerged, proposing additional factors and refinements to the models. Researchers have examined various drivers of stock returns, including firm-level quality characteristics (Novy-Marx, 2013; Asness, Frazzini and Pedersen, 2013). Other studies have proposed alternative frameworks, such as the q-factor model, which offers a slightly different approach to explaining stock returns (Hou, Xue & Zhang, 2015). Despite these developments, the validity and importance of certain factors in the original Fama-French models remain subject to discussion. In particular, the robustness and the economic relevance of the size factor (SMB) has been repeatedly questioned. This thesis contributes to this discussion by examining both a reproduced version of the original Fama-French five-factor model and a modified version in which the size factor is adjusted using different quality measures. It motivates why incorporating a quality adjustment to the size factor may increase its relevance and aims to test this empirically. It also assesses the relevance and explanatory power of the original model and evaluates the impact of the adjustment on the model and on the size factor itself. By testing the models over the 1972–2024 period, the analysis provides evidence on whether such an adjustment improves asset pricing performance relative to the original specification.

2 Literature Review and Theoretical Background

Since its introduction in 1993, the three-factor model by Fama and French has heavily influenced asset pricing literature. In the following years, it became a cornerstone of empirical finance, and estimating alphas with respect to the model's factors became standard practice in academic research. The three-factor model extends the CAPM by examining the role of firm size and the book-to-market equity ratio in explaining the cross section of average stock returns.

In their studies Fama and French reported that the market beta, defined as the slope of the regression of a stock's return on the market's return, gave little information about the stock's average returns. This finding challenged the CAPM's prediction that the market beta should be the dominant determinant of expected returns. To address this shortcoming, they proposed the addition of two new variables to the right-hand-side of the original CAPM equation proposed by Sharpe. They introduced the size factor (SMB or Small Minus Big), to capture the higher average returns of smaller firms over larger firms. In addition to size, they introduced the value factor (HML or High Minus Low), designed to capture the premium that firms with high-book to market equity ratios have over those with low ones.

Both factors were constructed from portfolios that sorted firms into groups based on their market capitalization and their book-to-market ratios. The SMB factor was calculated as the return on a portfolio of 'small' firms minus the return on a portfolio of 'large' firms, isolating the size premium. Similarly the HML factor was the return difference between a portfolio of value stocks (stocks with high book-to-market ratios) and a portfolio of growth stocks (stocks with low book to market ratios), isolating the value premium.

The results reported by Fama and French provided strong evidence supporting the

addition of those two new factors to the CAPM. They found that including the size and value factors improved the ability of the model to explain the variation in returns across diversified portfolios. In doing so, they argued that these factors could capture a part of systematic risk that was not captured by the market beta alone in the CAPM. Therefore the model extended the CAPM with the size (SMB: Small Minus Big) and value (HML: High Minus Low) factors.

However, in the following years many studies highlighted the limitations of the three-factor model. While it substantially improved the explanatory power of the CAPM, researchers observed that the alphas estimated under this framework were often statistically different from zero. This implied that the three-factor model was still incomplete, leaving part of the cross-section of stock returns unexplained. These critiques led to the idea that the model might require exposure to more than just three factors. Evidence from Novy-Marx (2013) and other related studies suggested that the model overlooked the part of the variation in average returns that was related to profitability and investment.

Following this idea, Fama and French (2015) proposed the introduction of two new factors to increase the performance of the three-factor model, thereby transforming it into a five-factor model. The new version of the asset pricing model incorporated namely the profitability (RMW: robust minus weak) factor and the investment (CMA: Conservative minus Aggressive) factor. In line with the construction process of the original factors, in the new model's equation RMW is defined as the difference between the returns on diversified portfolios of stocks with robust and weak profitability, while CMA is the difference between the returns on diversified portfolios of stocks of firms with low and high investment values, which were labeled conservative and aggressive firms by Fama and French. The five factor model equation became:

$$\begin{aligned}
R_{p,t} - R_{f,t} = & \alpha_p + \beta_{p,MKT}(R_{m,t} - R_{f,t}) + \beta_{p,SMB}(SMB_t) \\
& + \beta_{p,HML}(HML_t) + \beta_{p,RMW}(RMW_t) + \beta_{p,CMA}(CMA_t) + \epsilon_{p,t} \quad (1)
\end{aligned}$$

Just as with the three-factor model, the performance of Fama and French's new creation can be assessed by analyzing the estimated alphas of the regression. If the exposure to the five factors, represented by the Betas associated to each factor, captures all the variation in expected returns, then the intercept of the regression should not differ significantly from zero for any portfolio p .

While the introduction of the new factors increased the accuracy of the previous framework and marked an advancement over the three-factor model, several studies have pointed out that the five-factor model still retains some issues that weaken its explanatory power. Among these studies, Blitz, Hanauer, Vidojevic & van Vliet (2016) emphasizes that the two new factors, RMW and CMA, may lack robustness and have limited relevance in other asset classes besides common stocks. The same study also raises questions about the economic rationale of the two new factors.

In addition to these critiques, many other different papers have questioned other aspects of the model. In this thesis, I focus on the criticism related to the size factor.

Banz (1981) found that small stocks in the U.S. in terms of market capitalization, tend to have higher average returns than larger stocks. This effect did not get accounted for by the higher market beta of smaller firms. The same paper also reported that the size was concentrated more on very small firms while there was little difference in return between average sized and large firms. Moreover, it also added that it was unclear if size per se was responsible for the size effect or if it was merely a proxy for one or more unobserved factors correlated with size. The identification of the size effect and

the other factors also had a substantial impact on the investments market.

Despite being discovered many years prior to its inclusion in the five factor model , the size factor has since faced criticism for its weak empirical performance, particularly compared to the other factors in the model. Other major critiques to the size factor, studies have noted that the size premium appears to have weakened since its discovery, that it is disproportionately concentrated in January, and that its persistence over long horizons is questionable. All of these critiques have cast doubt on the relevance of the size premium within the five-factor model .

The objectives of this thesis are two: first to replicate the results and factors of the original five-factor model proposed by Fama and French, and second, to modify the construction of the size factor. Specifically, I examine whether incorporating firm quality as a control variable can enhance the relevance of the size factor, and improve the overall reliability of the model.

The idea that accounting for firm's quality could strengthen the performance of the model and be a meaningful adjustment for the size factor is supported by several studies such as 'Size Matters, If you control your Junk' (2018) and 'Quality minus Junk' (2019), which provide empirical and theoretical evidence as to why this kind of adjustment can be relevant. Intuitively, quality of a stock can be defined as a set of characteristics that, all else equal, can justify a higher price for a stock. These are essentially traits that investors should be willing to pay a premium for. Furthermore, since the size factor is based on market values, and market values themselves are influenced by these quality-related characteristics, these two variables (size and quality) are expected to be strongly connected.

Motivated by this relationship, I selected a set of accounting measures to classify firms into high-quality and low-quality firms, formed portfolios based on these rankings and used the resulting portfolios to adjust the construction process of the size factor

in the model. I then tested both the original version of the Fama French five-factor model and the quality-adjusted version, evaluating the results to determine whether the replication was accurate and the extent to which the adjustment to the SMB factor improved its relevance and the overall performance of the model.

3 Data and Methodology

In this section I describe the data used in this study, both in the replication of the Fama-French five-factor model factors, and for the part of the analysis where I reassess the size factor by controlling for quality measures.

3.1 Replicating the Fama-French Five-Factor model

Monthly stock return data come from the Center for Research in Security Prices (CRSP) database and includes all available common stocks on CRSP between January 1972 and December 2024. To ensure comparability with the Fama-French (2015) methodology and the original sample, only common stocks were included (sharecodes 10 or 11), excluding other security types such as preferred stocks to avoid potential distortions. Furthermore, the sample is restricted to U.S. stocks listed on the NYSE, AMEX or NASDAQ. Delisted stocks were treated in the following way: if a value for the delisting return was available, it was assigned to the stock; if unavailable, a value of -30% was applied, following Shumway (1997). Excess return were calculated using as risk free rate the one-month U.S. Treasury bill Rate, as provided by the Fama-french data library on Ken French's website.

Estimates of the original five factors were downloaded from Ken french's website, which provides monthly observations from January 1964 to December 2024. These were then adjusted to match the sample used in this thesis ,which spans from 1972 to

2024. The replication of the model was then tested against the twenty-five 5x5 test portfolios which are available on the Ken French's Data Library. These portfolios represent the intersection of five portfolios formed on size and five formed on the book-to-market equity ratio. Unlike the 2x3 sort, the size breakpoints for these portfolios are based on NYSE market equity and book-to-market equity ratio quintiles for each year. The portfolios are labeled from s1b1 through s5b5, where the first number refers to the size quintile (1 = smallest firms, 5 = largest) and the second one refers to the book-to-market equity ratio quintile (b1 = lowest, b5 = highest). Thus, each label uniquely identifies a portfolios' position along the size and book-to-market grid. To ensure compliance with the rest of the methodology, these portfolios are built yearly in June.

Financial Data was collected from the Compustat North America database, which includes U.S. equities financials in the sample period January 1972-December 2024. The accounting data were linked to the stock returns data via company identifiers to ensure accurate matching between the two datasets. To avoid duplication from firms with multiple filings within the same year, only the final annual observation for each company was included .

In order to construct the factor portfolios, the accounting data were first cleaned and firm-level measures were created. The factors used in the original Fama-French 5 factor model are based on four different accounting variables: book value of equity, operating profitability, investments and market capitalization. Since accounting data are usually assumed to be publicly available 3-6 months after the fiscal year-end, factor portfolios are formed each June using the most recent available accounting data. Therefore to ensure a compatibility between market and accounting data, a reference date in June of each year was created to join the two databases. Market data (stock returns and market capitalization) for months January-June are matched with accounting data filed in the previous year, while market data for months July-December are

matched with accounting data filed in the current year.

The Book value measure was constructed as follows:

$$\begin{aligned} \text{Book Value} = & \text{Stockholder's Equity} + \text{Deferred Taxes and Investment Tax Credits} \\ & - \text{Preferred Stocks} \end{aligned} \quad (2)$$

Furthermore only observations that presented a positive Book value were included in the sample to avoid problems with Book-to-Market ratios during the construction of the factor portfolios, and to remain consistent with standard practice. Then, following the standard practice regarding the June lag, the following formula was used to construct the Book-to-Market ratios used later to form the factor portfolios:

$$\frac{\text{Book}}{\text{Market}} = \frac{\text{Book Value}}{\text{Market Equity Value}} \quad (3)$$

The Operating Profitability measure was calculated as:

$$\text{Operating Profitability} = \frac{\begin{aligned} & \text{Total Revenues} - \text{COGS} - \text{Interest Expenses} \\ & - \text{Selling and Administrative Expenses} \end{aligned}}{\text{Stockholder's Equity}} \quad (4)$$

To ensure a higher consistency, missing data for Interest expenses and for Selling and Administrative expenses, were assigned a value equal to zero, under the assumption that these costs are unreported. Furthermore, following the standard Fama-French procedure, only companies reporting non-null values for total revenues and COGS were admitted in the sample. The Investment measure was constructed as the annual growth

in total assets, with the formula:

$$\text{Investment} = \frac{\text{Total Assets}_t - \text{Total Assets}_{t-1}}{\text{Total Assets}_{t-1}} \quad (5)$$

In this case, only companies that presented a positive value of total assets were retained in the sample, to avoid distortion and to ensure compatibility with the rest of the analysis. Finally, for market capitalization, the value provided by the CRSP database was used directly.

3.2 Sorting Process

In order to sort the stock returns observations with respect to their size, the median market capitalization for the sample is calculated each June. Each observation is then assigned to a size portfolio and labeled 'B' (Big) if its market capitalization is greater than the median and 'S' (Small) otherwise. To closely follow the original Fama French procedure, the median market capitalization is calculated only accounting for stocks listed on the NYSE. Since NYSE stocks tend to be larger, this approach prevents the median from being too low, which would ultimately result in most stocks being classified as Big.

To sort the stock returns by book-to-market ratio, breakpoints are created to classify each observation. Such breakpoints are the 30th and 70th percentiles for book-to-market ratio calculated in June. Following the previous rationale, the breakpoints are calculated only on NYSE stocks. Each observation is then classified as 'H' (High) if its BM ratio is higher than the 70th percentile, 'L' (Low) if it's lower than the 30th percentile, and 'M' (Medium) in the other cases.

The robustness sort based on operating profitability follows a similar procedure. Breakpoints are determined using the 70th and 30th profitability percentiles of operating

profitability for NYSE stocks. Observations are then classified based on the breakpoints into 'R' (Robust) if their operating profitability is greater than the 70th percentile, 'W' (Weak) if it is lower than the 30th percentile and 'M' (Medium) otherwise.

Finally, the investment sort is based on the investment measure, defined as growth in total assets. Investment breakpoints were calculated using the same 70th and 30th procedure in June, and observations are then categorized as 'C' (Conservative), 'N' (Neutral) and 'A' (Aggressive).

3.3 Factor Creation

Following the Fama French methodology to build the factors, stocks are sorted using a 2x3 sort, resulting in 6 value-weighted portfolios based on size and book to market ratio, 6 value-weighted portfolios based on size and operating profitability and 6 value-weighted portfolios based on size and investment.

The SMB factor is calculated as the average return on all small stock portfolios minus the average return on all big stock portfolios, namely three sub-portfolios are created:

$$\begin{aligned} \text{SMB}_{B/M} = & \frac{1}{3}(\text{Small High} + \text{Small Medium} + \text{Small Low}) \\ & - \frac{1}{3}(\text{Big High} + \text{Big Medium} + \text{Big Low}) \end{aligned} \quad (6)$$

$$\begin{aligned} \text{SMB}_{Op} = & \frac{1}{3}(\text{Small Robust} + \text{Small Medium} + \text{Small Weak}) \\ & - \frac{1}{3}(\text{Big Robust} + \text{Big Medium} + \text{Big Weak}) \end{aligned} \quad (7)$$

$$\begin{aligned} SMB_{INV} = & \frac{1}{3}(\text{Small Conservative} + \text{Small Neutral} + \text{Small Aggressive}) \\ & - \frac{1}{3}(\text{Big Conservative} + \text{Big Neutral} + \text{Big Aggressive}) \end{aligned} \quad (8)$$

$$SMB = \frac{1}{3}(SMB_{B/M} + SMB_{Op} + SMB_{INV}) \quad (9)$$

The HML factor is then calculated as the average return on the two high book-to-market portfolios minus the average return on the two low book-to-market portfolios

$$HML = \frac{1}{2}(\text{Small High} + \text{Big High}) - \frac{1}{2}(\text{Small Low} + \text{Big low}) \quad (10)$$

The RMW factor represents the average return on the two robust profitability portfolios minus the average return on the two weak profitability portfolios:

$$RMW = \frac{1}{2}(\text{Small Robust} + \text{Big Robust}) - \frac{1}{2}(\text{Small Weak} + \text{Big Weak}) \quad (11)$$

The CMA factor is constructed as the average return on the two conservative investment portfolios minus the average return on the two aggressive ones:

$$CMA = \frac{1}{2}(\text{Small Conservative} + \text{Big Conservative}) - \frac{1}{2}(\text{Small Aggressive} + \text{Big Aggressive}) \quad (12)$$

Finally, the MKT factor represents the excess return on the market, computed as the value-weighted return of all U.S. stocks in the CRSP database and using as risk-free rate the one-month Treasury bill rate.

Tables 1, 2 and 3 report descriptive statistics regarding the number of constituents

in the different factor portfolios.

Table 1: Size–Value Portfolio Constituents (1972–2024)

Size	Value	Mean	Min	Max
Small	High	1002	515	1732
Small	Medium	929	422	1526
Small	Low	729	174	1135
Big	High	136	80	217
Big	Medium	293	235	391
Big	Low	364	247	644

Table 2: Size–Profitability Portfolio Constituents (1972–2024)

Size	Profitability	Mean	Min	Max
Small	Robust	486	227	834
Small	Neutral	848	400	1335
Small	Weak	1326	501	2211
Big	Robust	300	214	389
Big	Neutral	320	222	411
Big	Weak	174	106	338

Table 3: Size–Investment Portfolio Constituents (1972–2024)

Size	Investment	Mean	Min	Max
Small	Conservative	972	487	1465
Small	Neutral	805	363	1353
Small	Aggressive	884	294	1535
Big	Conservative	169	112	255
Big	Neutral	335	294	412
Big	Aggressive	291	206	447

3.4 SMB Quality Adjustment

To build the quality adjustment to the smb factor I first calculated three different accounting measures for every company, namely asset turnover, defined by the formula:

$$\text{Asset Turnover}_t = \frac{\text{Revenue}_t}{\text{Total Assets}_t} \quad (13)$$

Interest coverage:

$$\text{Interest Coverage}_t = \frac{\text{EBIT}_t}{\text{Interest expenses}_t} \quad (14)$$

and gross profitability:

$$\text{Gross Profitability}_t = \frac{\text{Revenues}_t - \text{COGS}_t}{\text{Total Assets}_t} \quad (15)$$

Typically a higher asset turnover suggests that a firm is efficient at generating revenues from its assets. Conversely, a lower asset turnover could indicate operational inefficiency therefore suggesting a lower quality firm. Similarly, a higher interest coverage ratio shows that a firm is able to meet its debt obligations, which in turns helps reducing financial risk. This characteristics increases the safety of a firm which can be associated to a higher quality firm. Lastly, a higher gross profitability can signal the strength of the firm's core business, making the company more resilient and therefore signaling a higher quality firm. To avoid distortions in calculating the ratios, only observations that presented non-null values for total assets and interest expense were included in the sample. Additionally, the same timing adjustment as the original five factor model process were made, creating a reference date in June used to align accounting and market data and maintain consistency with standard practice.

The observations for each measure were then converted into percentile ranks within the year and then averaged into a quality score. I then used the sample to create yearly breakpoints to help with the classification of the observations. Following the same procedure used for the construction of the other factors, the breakpoints were formed at the 30th and 70th percentiles of the quality score and only stocks listed on the NYSE were included in the calculations.

I then classified each observation into three different categories based on its quality

score: high quality if its score was higher than the 70th percentile of the quality scores, low quality if it was lower than the 30th percentile and medium quality otherwise. To incorporate the adjustment in the smb factor, six value weighted portfolios based on size and quality were created using the same 2x3 sorting procedure as the other factors. Then, the sub-portfolio $SMB_{Quality}$ was created, as:

$$SMB_{Quality} = \frac{1}{3}(\text{Small High Quality} + \text{Small Medium Quality} + \text{Small Low Quality}) - \frac{1}{3}(\text{Big High Quality} + \text{Big Medium Quality} + \text{Big Low Quality}) \quad (16)$$

Finally, to include this portfolio into the construction process of the size factor, the SMB was modified to:

$$SMB = \frac{1}{4}(SMB_{B/M} + SMB_{Op} + SMB_{INV} + SMB_{Quality}) \quad (17)$$

This approach ensures compatibility across the accounting measures, while giving equal weight to each quality dimension used in the construction of the factor. Table 4 reports descriptive statistics regarding the number of constituents in the size-quality factor portfolios.

Table 4: Size–Quality Portfolio Constituents (1972–2024)

Size	Quality	Mean	Min	Max
Small	High Quality	237	121	417
Small	Medium Quality	1489	826	2171
Small	Low Quality	291	121	417
Big	High Quality	73	54	106
Big	Medium Quality	539	419	676
Big	Low Quality	62	40	108

4 Statistical analysis

4.1 Model's Reproduction Accuracy

This thesis has three main objectives: to provide an accurate reproduction of the Fama-French five-factor model over the sample period January 1972-December 2024, to test the model against the test portfolios available on Ken French's website, and finally to modify the construction process of the SMB factor to examine the impact said adjustment has on the factors and on the model's overall performance. To assess the reproduction, I first compared the performance of the five factors I estimated with the original factors available on Ken French's data library. Table 5 and 6 present descriptive statistics for both sets of factors over the sample period. Both tables include a number of observations equal to 630, covering the sample period June 1972 to December 2024. Fama-French factor portfolios are created in June of each year, this makes it so that the months previous to June in the first year are not part of the sample period for the factors observations.

Table 5: Reproduction of the Model: Factor's descriptive Statistics

	MKT	SMB	HML	CMA	RMW	SMB-Quality
Observations	630	630	630	630	630	630
Mean	0.0071	0.0019	0.0023	0.0020	0.0027	0.0023
Std	0.0462	0.0302	0.0313	0.0213	0.0187	0.0238
Min	-0.2247	-0.1140	-0.1517	-0.1183	-0.0582	-0.2258
Max	0.1657	0.1492	0.1479	0.1094	0.0791	0.1630

Table 6: Original Fama-French Model: Factor's descriptive Statistics

	MKT FF	SMB FF	HML FF	CMA FF	RMW FF
N. observations	630	630	630	630	630
Mean	0.0062	0.0013	0.0030	0.0029	0.0028
Std	0.0458	0.0302	0.0309	0.0230	0.0204
Min	-0.2319	-0.1554	-0.1383	-0.1895	-0.0708
Max	0.1610	0.1846	0.1286	0.1305	0.0901

The highest mean values of 0.00713 and 0.00621 are reported by the MKT factor in both tables. This indicates that the exposure to the market factor contributes substantially to the monthly return of the stocks in the sample. On the other hand, the lowest mean values are reported by the SMB factor in both tables with a mean of 0.00192 and 0.00136 for the reproduced and original factors respectively. The remaining factors display mean values that are very close to each other. From the same table it is also noticeable that the factor with the highest variation, represented by the standard deviation measure, is the market factor. It presents a standard deviation of 0.0462 and 0.0458 in the reproduction and in the original model respectively. Table 7 reports the differences between mean values, maximum and minimums for each factor pair, calculated by subtracting the original fama french factor from the factors I estimated. It is noticeable that the largest discrepancies between the mean value of a factor are observed for the MKT and RMW factors, with differences of 0.00091 and 0.0008 respectively. Conversely, the smallest difference occurs for the CMA factor with a value of -0.00009. This suggests that the Investment factor aligns most closely with its counterpart in terms of average value, while the market and profitability factors show the largest deviations from the originals.

Table 7: Descriptive Statistics for Factor Differences

	MKT diff	SMB diff	HML diff	CMA diff	RMW diff
Mean	0.00091	0.00055	-0.00062	-0.00089	-0.0000
Min	-0.0615	-0.0391	-0.0713	-0.0860	-0.0441
Max	0.0519	0.0413	0.0495	0.0880	0.0568

To better understand how well each factor in my reproduction of the model tracks its Fama-French counterpart, I compared the time series of each factor over the whole sample period.

Figure 1 shows the monthly time series from January 1972 to December 2024 for the original Fama-French factors alongside the time series of the five factors constructed

in this study. The graphs clearly indicate that all of the factors closely follow their counterparts, both during stable periods and in times of large fluctuations, as shown by the two lines closely mirroring each other in the figure.

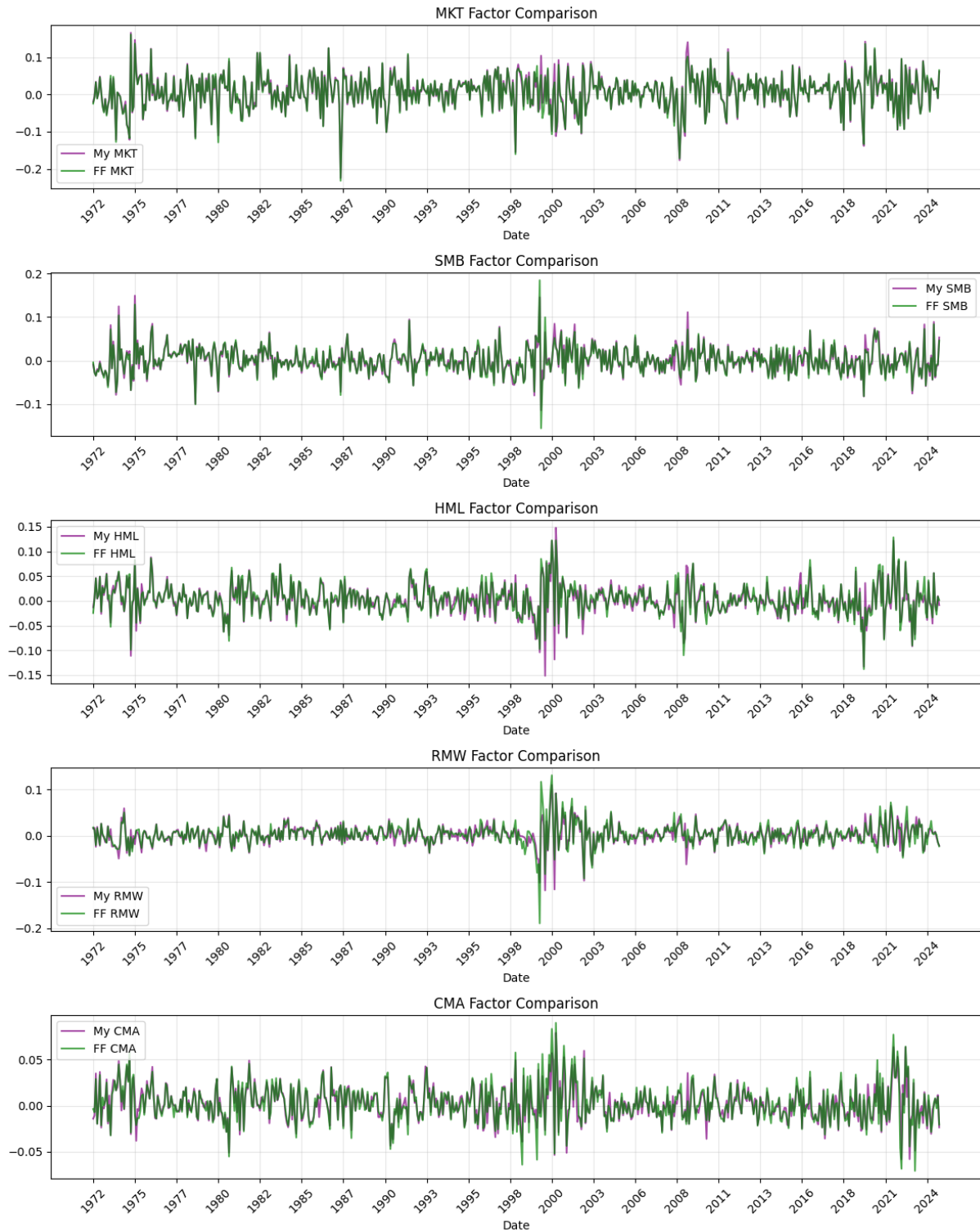


Figure 1: Monthly time series (January 1972- December 2024) comparison between estimated factors and original factors from Ken French's websites

Since the time series can only give a glimpse of how accurately the factors are tracking their objective, I also calculated all the correlations between each factor and constructed a heatmap to visualize them all in the same graph. The heatmap in figure 2 shows the correlation between each original factor from Ken French's website and the ones I constructed and also the covariances between all the different factors. The graph confirms the conclusion that was suggested by the time series comparison. Each factor shows a correlation value higher than 90% with its respective counterpart. It is also observable that the factors that report the highest correlations are the market (MKT) and size (SMB) factors with values of 99.1% and 98.8%, while the lowest is reported by the profitability (RMW) factor with a value of 90.2%.

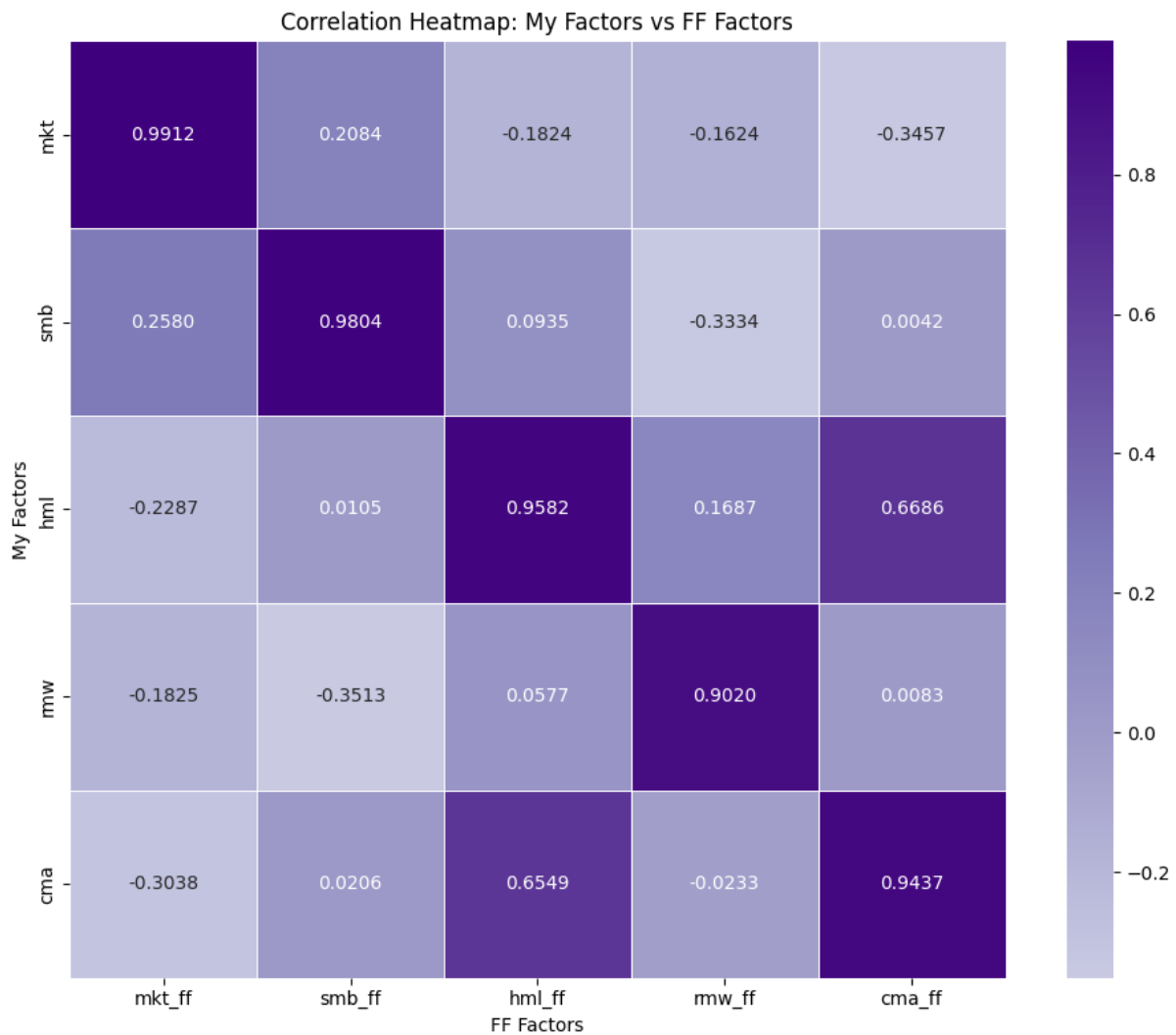


Figure 2: Correlation heatmap between factors

To further confront the factors I created boxplots for each factor and confront them with their Fama-French's counterpart. This is done to highlight how similar the distributions of each factors are to the original available on Ken French's website. The comparison between the boxplots in figures 3, 4, 5, 6, 7 shows how the two distribution behave compared to each other.

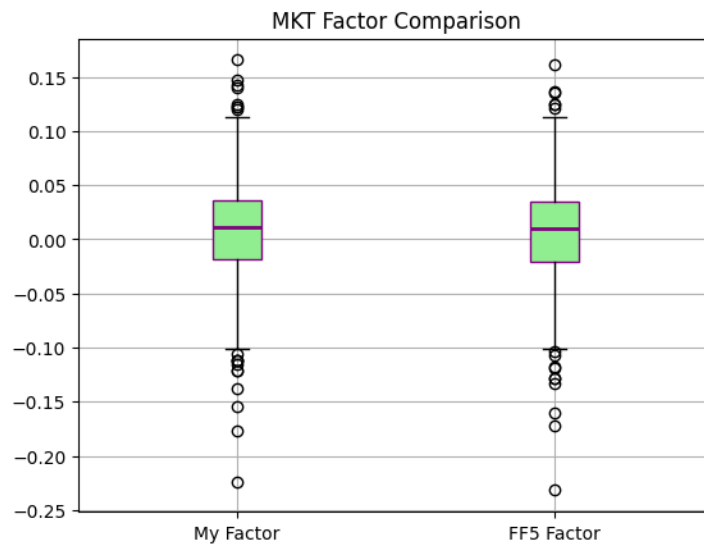


Figure 3: MKT Boxplots comparison

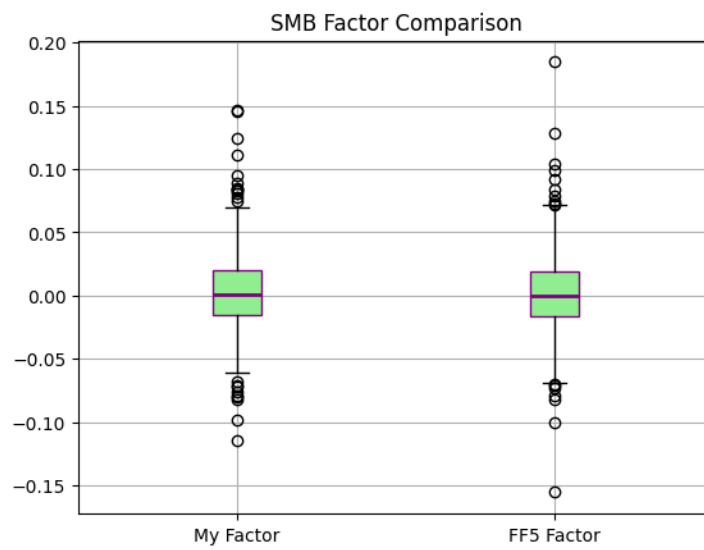


Figure 4: SMB Boxplots comparison

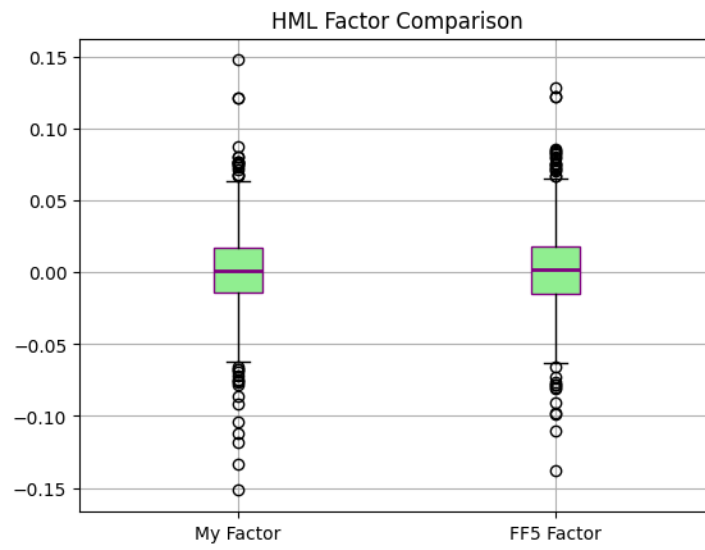


Figure 5: HML Boxplots comparison

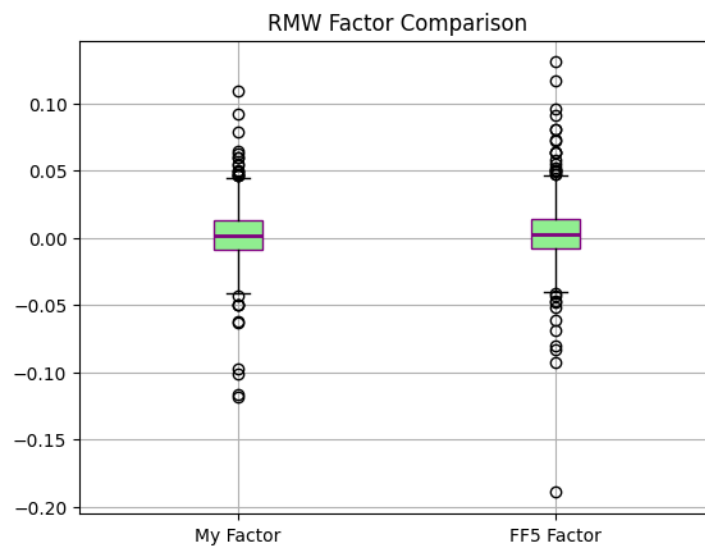


Figure 6: RMW Boxplots comparison

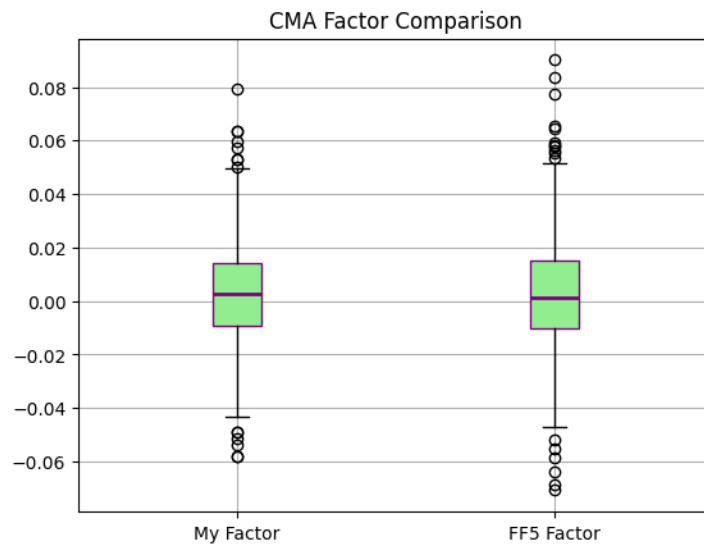


Figure 7: CMA Boxplots comparison

Finally, a last test of the similarity between the two set of factors can be done through the Kolmogorov-Smirnov test (KS test). This test statistic is used to compare the cumulative distribution functions of two datasets, in this case to compare each pair of factors. The KS test statistics takes value in the range 0-1 based on how similar are the two cumulative distribution function. Values closer to 0 indicate a high similarity and values closer to 1 signalize the opposite conclusion. As shown in figure 8, the factors that report the lowest KS statistics are the MKT (market) factor and the SMB (size), with KS statistics of 0.0190 for the former and 0.0206 for the latter. Although the other factors show higher statistics, with the CMA (investment) factor showing the highest value of 0.0349, all of the test statistics show values in the range 0.019-0.0349, which are very close to 0. These confirm that the distributions of the reproduced factors closely match the original ones. Furthermore, the p-values for all KS test statistics are above the 5% significance level, indicating that the null hypothesis that the samples come from the same distribution cannot be rejected.

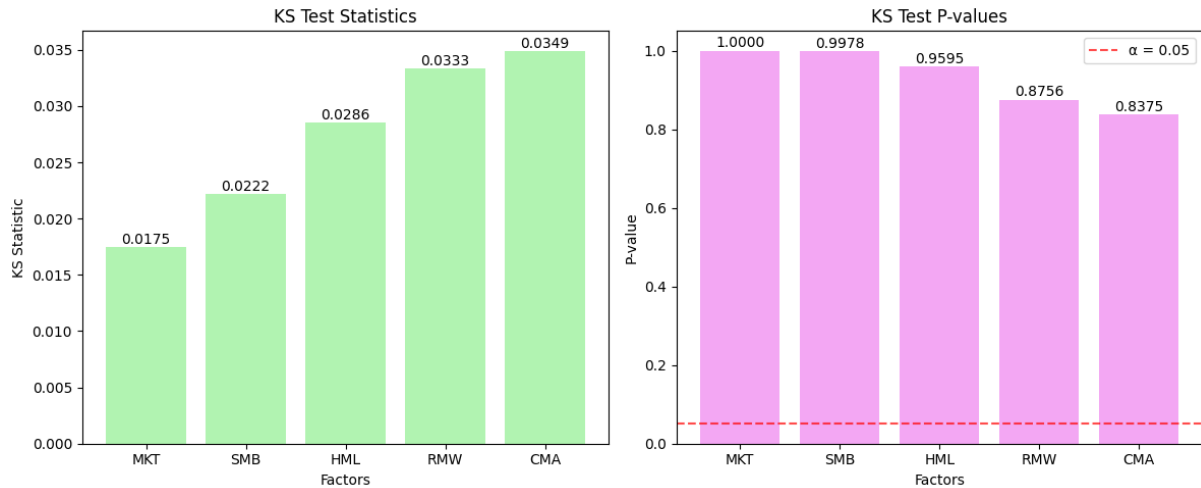


Figure 8: Histograms showing the KS test statistics and their associated p-value for each factor pair

All of these tests support the conclusion that the reproduction of the model has been successful, as all factors exhibit high correlation and similar distributions relative to their original Fama-French counterparts. Having verified that the model has been reproduced, I now focus on the size factor. First, in order to assess the relevance of said factor, I considered the various critiques that the SMB factors has faced over the years and that many studies have presented. As previously noted, in the analysis the size factor has reported the lowest average value among all the factors, both in my reproduction of the model and in the original Fama-French model. This provides supporting evidence for the critiques found literature, suggesting that the size factor is less influential then the others factors in the model. Furthermore, from tables 6 and 5 it is evident that the size factor places itself as one of the most volatile factors, due to its standard deviation and to the range between its median and its maximum and minimum values. Another common concern raised from different studies about the size factor, is the so-called 'January effect'. That indicates that the size premium is

disproportionately concentrated in the first month of the year,. In order to test if these concerns are empirically based, I separated the size factors observations into January and non-January observations. I then calculated the mean and standard deviations for both classes, which are reported in Table 8. For the reproduction of the model, the SMB factor has a mean of 0.0133 in January while an average value of merely 0.0009 in the other months. Similarly, for the original Fama-French size factors the values are 0.0113 and 0.0005, respectively. From these results it appears that the majority of the size premium does indeed occur to January, both for the reproduction of the model and for the original Fama-French factors.

Table 8: Descriptive Statistics for January Effect

	January	Non January	January FF	Non January FF
Observations	52	578	52	578
Mean	0.0133	0.0009	0.0113	0.0005
Std	0.0410	0.0290	0.0385	0.0292

Furthermore, Table 9 reports the average difference between January and non-January SMB values, as well as the associated p-values. Both models show a mean difference value significantly different from zero, a conclusion that is also supported by the low p-values associated to it. This ultimately confirms the empirical presence of the January effect over the sample period.

Table 9: Descriptive Statistics for January Effect, including comparison with post-quality adjustment results

	January Vs Non January	January Vs Non January FF	January Vs Non January Post quality adjustment
Average Difference	0.0124	0.0109	0.0065
T-Stat	2.8533	2.4898	1.8800
P-value	0.0045	0.0130	0.0606

To examine the relative contribution of January to the annual size premium, table 10 presents descriptive statistics for the January contribution, calculated for each year

as:

$$\text{Contribution}_{January} = \frac{SMB_{January}}{SMB_{\text{Rest of the Year}}} \quad (18)$$

Table 10: January contribution to yearly SMB factor

	Model Reproduction	Original Fama-French	Post-Quality Adjustment
n.Years	41	41	37
Average Contribution	21.0%	17.8%	13.9%
lower bound	-43.7%	-66.2%	-49.8%
upper bound	99.0%	89.5%	76.1%

To avoid distortion with extreme years, only observations where the January contribution didn't exceed +100% or -100% of the yearly size factor observation were included in the calculations. On average, the January contribution is significantly more than what should be expected from just one month out of the year. Theoretically if we assume an equal distribution of the size premium across the months in the year, the single month contribution shouldn't exceed 8.3%. In both the reproduction and the original model, the January contribution values exceed that threshold, indicating the existence of the January effect. Figure 9 shows the time series of the January contribution to the yearly SMB factor over the sample period. By examining the time series and the range defined by upper and lower bound in table 10 it's possible to understand how the January effect behaves over the sample period. It is evident that for most years in the sample, the January contribution surpasses the expected threshold. Furthermore, not all of the values shown are positive, indicating that in some years the negative January effect contributed to reducing the overall yearly size premium. Taken together, these findings lead to the conclusions that the size premium majorly reside in January, confirming the presence of the 'January Effect'.

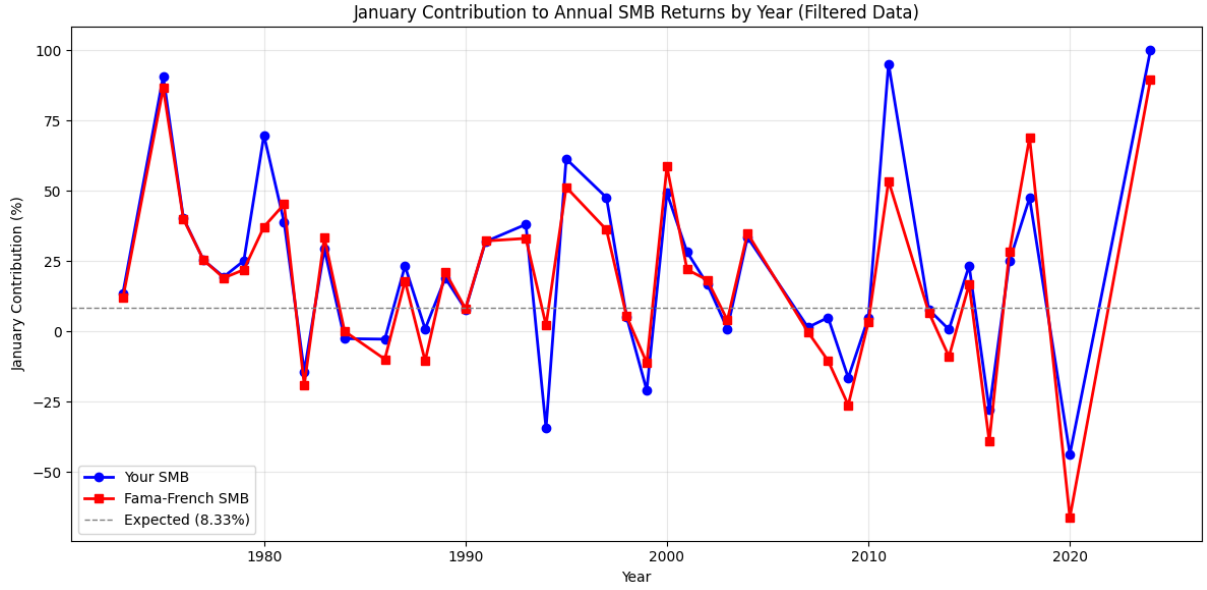


Figure 9: Shows the time series of January contribution to yearly Size premium

4.2 Model's explanatory power Pre-Adjustment

After establishing that the reproduced model and factors are a close approximation of the original one, I now turn to testing the model itself, to evaluate its explanatory power and to better understand the critiques often raised against it. To assess the validity of the model and of its factors, I tested both the reproduced and original Fama-French five factor models against the 25 5x5 size-book-to-market test portfolios available on Ken French's data library. The regression framework follows the standard regression:

$$\begin{aligned}
 R_{p,t} - R_{f,t} = & \alpha_p + \beta_{p,MKT}(R_{m,t} - R_{f,t}) + \beta_{p,SMB}(SMB_t) \\
 & + \beta_{p,HML}(HML_t) + \beta_{p,RMW}(RMW_t) + \beta_{p,CMA}(CMA_t) + \epsilon_{p,t} \quad (19)
 \end{aligned}$$

where the element $R_{p,t} - R_{f,t}$ denotes the excess return on portfolio p for month t . The coefficients $\beta_{p,Factor}$ measure the portfolio's exposure to each factor present in the model, while α_p represents the portion of each test portfolios' excess return that does not get captured by the model. Finally the element $\epsilon_{p,t}$ is the error term for portfolio p in

month t . This setup allows me to assess the model's ability to explain variation in excess returns. The most relevant outputs are the estimated alphas, their corresponding t-statistics, and the adjusted R^2 . The alpha reflects the part of each test portfolios' excess return that doesn't get explained by the model, while its t-statistic indicates whether this residual is statistically different from zero. The adjust R-squared captures the proportion of the variation in returns accounted for by the factors, indicating the overall fit of the model and its explanatory power.

Table 11 reports the estimated alphas across the 25 test portfolios. Overall, the model achieves a high level of explanatory ability, with an average adjusted R^2 of 0.9051. This indicates that, on average, the five factors explain more than 90% of the variation in excess returns across the set of test portfolios, which is consistent with the performance originally documented by Fama-French (2015). Turning to the alphas, their magnitudes are generally small with an average of -0.0009 and with most lying in a narrow range around zero. This suggests that in many cases the model does a good job at pricing the test portfolios. However, several alphas remain statistically significant at the 5% level, which can be seen by examining their p-values and t-statistics. For instance, portfolio s1b1 shows a high negative alpha of -0.0049 with a t-statistic of -4.62 and a very small p-value. Similar results are also displayed by other test portfolios in the analysis such as portfolios s2b1, s4b2, s4b3 and s4b5. This means that, although on average the model seems to perform well, it underestimates returns for certain portfolios, particularly for those at the extremes of the size and book-to-market spectrum.

By contrast, some portfolio show alphas that are not statistically different from zero. A notable example is portfolio s2b4 with an alpha of -0.000041, a t-statistic of -0.07 and a p-value of 0.94. These results indicate that the model prices this portfolios almost perfectly. Similar results are reported by portfolios s5b1 and s1b2. This illustrates that

Table 11: Regression results: portfolio alphas, adjusted R^2 , t-statistics, and p-values

Portfolio	Alpha	Adj. R^2	Alpha t-stat	Alpha p-value
s1b1	-0.00492	0.8978	-4.6244	0.0000
s1b2	0.00037	0.8971	0.3964	0.6919
s1b3	-0.00064	0.9383	-1.0254	0.3055
s1b4	0.00096	0.9219	1.4273	0.1539
s1b5	0.00079	0.8800	0.8717	0.3837
s2b1	-0.00222	0.9229	-2.6346	0.0086
s2b2	-0.00048	0.9372	-0.7628	0.4458
s2b3	-0.00074	0.9355	-1.2627	0.2071
s2b4	-0.00004	0.9336	-0.0703	0.9439
s2b5	-0.00100	0.9387	-1.5267	0.1273
s3b1	-0.00123	0.9255	-1.6362	0.1022
s3b2	-0.00048	0.9183	-0.7225	0.4702
s3b3	-0.00123	0.9129	-1.9521	0.0513
s3b4	-0.00069	0.9170	-1.1034	0.2702
s3b5	-0.00062	0.8967	-0.7696	0.4418
s4b1	0.00016	0.9102	0.2144	0.8302
s4b2	-0.00162	0.9052	-2.3949	0.0169
s4b3	-0.00164	0.8967	-2.3784	0.0176
s4b4	-0.00048	0.8828	-0.6631	0.5074
s4b5	-0.00172	0.8791	-2.0168	0.0441
s5b1	-0.00005	0.9491	-0.1138	0.9094
s5b2	-0.00107	0.8860	-1.6754	0.0943
s5b3	-0.00099	0.8591	-1.4266	0.1541
s5b4	-0.00293	0.8909	-4.3339	0.0000
s5b5	-0.00080	0.7942	-0.7343	0.4630
Average	-0.0009	0.9051	-1.2368	0.3057

while the model is broadly effective, its accuracy varies substantially depending on the portfolio's position in the size–book-to-market grid.

The results are visualized in figure 10 to provide a better representation. From the graphs it's clear that the model has performed best on the two test portfolios s2b4 and s5b1, since they show the lowest alpha across all the 25 test portfolios. On the other hand, the two portfolios on which the model has the worst performance are portfolios s1b1 and s5b4.

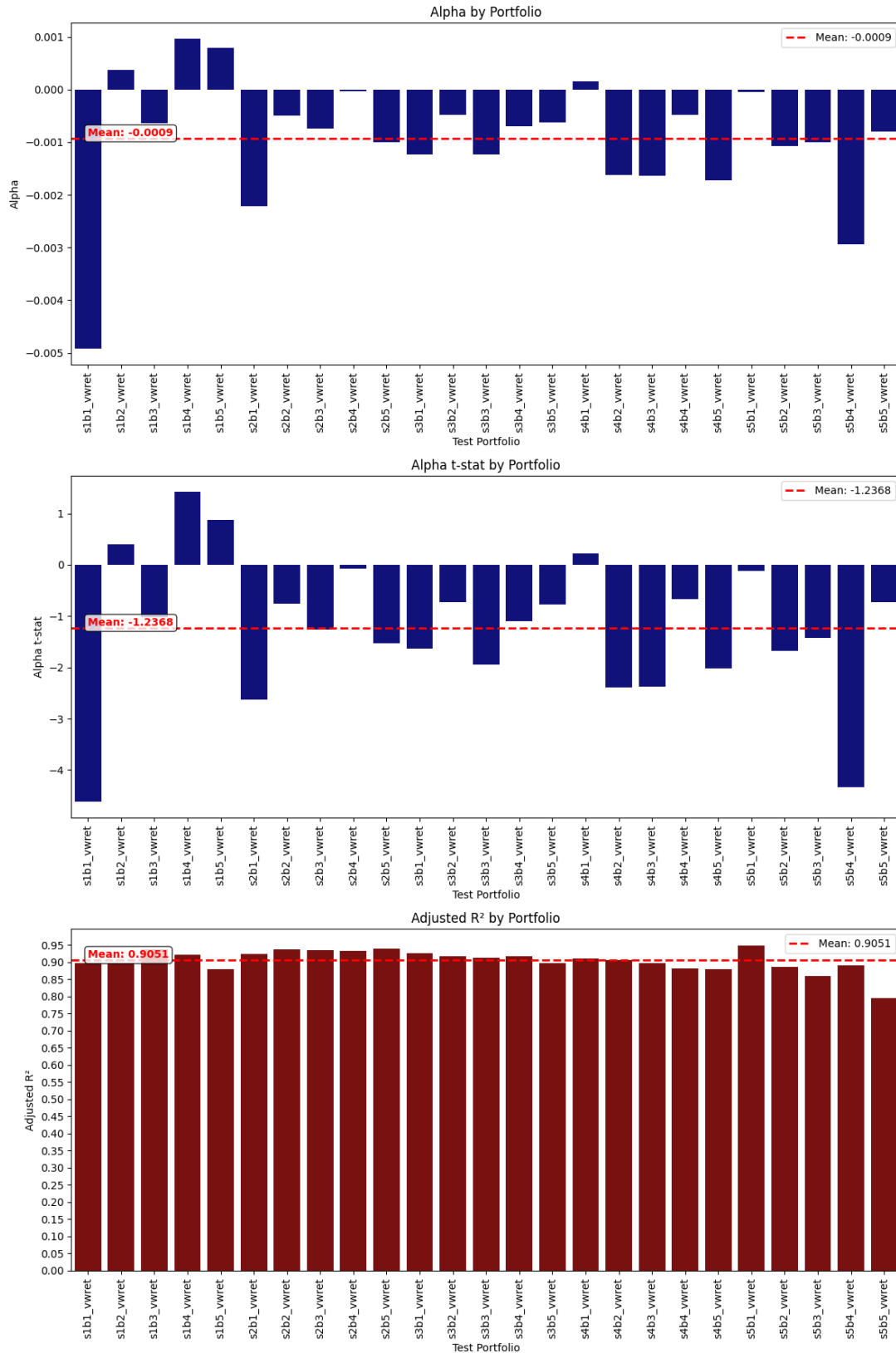


Figure 10: Reports alphas, alpha's t-statistics and adjusted R^2 of the regression used to test the estimated model against the 25 5x5 test portfolios

4.3 Performance After SMB Quality Adjustment

Having analyzed how the model performs in its original form, I now turn to the quality adjusted variant. The objective is to evaluate the impact of the quality adjustment on the smb factor and on the overall model. The last column of table 5 reports descriptive statistics for the size factor post quality adjustment. The results indicate that the size factor mean has increased from 0.0019 to 0.0023 after accounting for the quality adjustment, corresponding to a monthly increase of 21%. Comparing the factor pre and post adjustment it is also evident that, while the introduction of the quality adjustment had the effect of decreasing the standard deviation from 0.03029 to 0.023848, it also increased both the upper and lower bound of the size premium. The correlation between the pre- and post-adjustment size factors is 0.9098, which is lower than the correlation observed between the reproduced and original size factors. This outcome is consistent with expectations, as the introduction of the quality adjustment alters the construction process of the reproduced factor, therefore decreasing the extent to which it tracks the original one. At the same time, the result suggests that the adjusted factor captures a dimension of returns that is not fully reflected in the original one.

Examining the January effect for the size factor after the quality adjustment, table 12 shows that the adjustment substantially reduces the January effect. The average size premium in January decreased from 0.0133 pre-quality adjustment to 0.0083 post-adjustment, while the average non-January size premium rose from 0.0009 to 0.0018. Similarly, the average difference between January and non-January size premiums declined from 0.0124 to 0.0065, and the associated p-value increased from 0.0045 to 0.0606. Since this p-value now exceeds the 5% significance level, we can conclude that the difference is no longer statistically different from zero. These results indicate that the quality-adjusted size factor shows now a substantially reduced January effect, with no meaningful discrepancy between January and non-January average size premiums.

Table 12: Descriptive Statistics for January Effect Post-Adjustment

	January	Non January
Observations	52	578
Mean	0.0083	0.0018
Std	0.0319	0.0229

The average contribution of January to the yearly SMB factor post-adjustment is calculated following the same procedure as the pre-adjustment, excluding the years in which the January contribution fell outside the range $[-100\% +100\%]$. Figure 11 illustrates the time series of the average January contribution to the yearly size premium, after accounting for the quality adjustment.

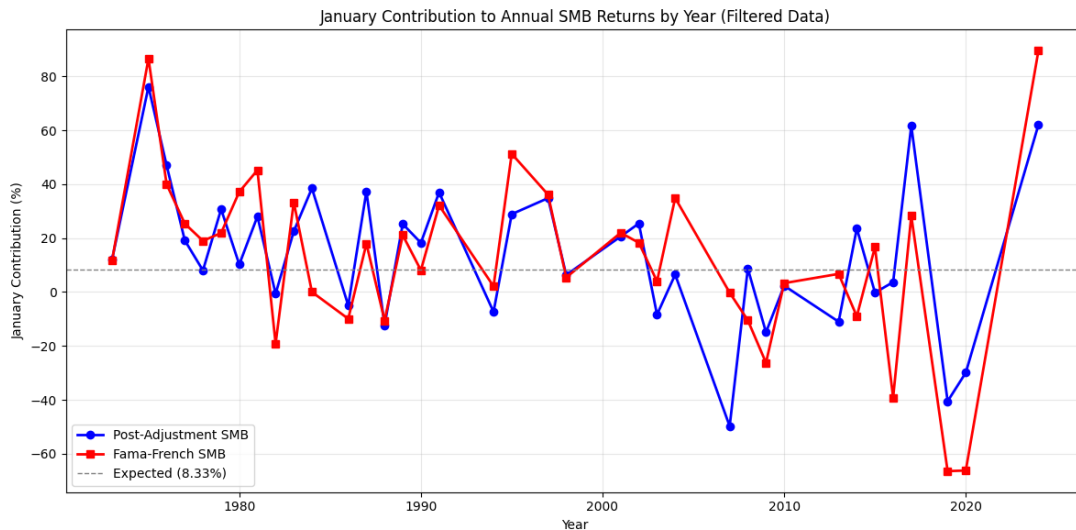


Figure 11: Shows the time series of January contribution to yearly Size premium, Post-quality adjustment

By comparing figure 9 11, it is apparent that the quality adjustment reduces the variability of the January contribution, creating lower peaks while allowing for periods of relatively high volatility. This observation is also supported by the data in table 10, where the lower bound has become slightly lower going from -43.7% to -49.8% , while the upper bound has significantly decreased from 99.0% to 76.1% post-adjustment.

Turning to the model as a whole, table 13 reports the estimated alphas across the

25 test portfolios. Comparing table 13 with table 11, it is evident that, while the average alpha across the test portfolios remains unchanged, the overall explanatory power of the model slightly declines, as the average adjusted R^2 decreases from 90% to 88.9%.

Table 13: Regression results (post-quality adjustment): portfolio alphas, adjusted R^2 , t-statistics, and p-values

Portfolio	Alpha	Adj. R^2	Alpha t-stat	Alpha p-value
s1b1	-0.00507	0.8646	-4.1210	0.0000
s1b2	0.00025	0.8660	0.2424	0.8085
s1b3	-0.00075	0.9147	-1.0288	0.3039
s1b4	0.00081	0.8892	0.9997	0.3178
s1b5	0.00057	0.8625	0.5825	0.5604
s2b1	-0.00233	0.9032	-2.4624	0.0140
s2b2	-0.00069	0.9261	-0.9976	0.3188
s2b3	-0.00087	0.9236	-1.3611	0.1739
s2b4	-0.00011	0.9061	-0.1683	0.8663
s2b5	-0.00111	0.9111	-1.4104	0.1589
s3b1	-0.00145	0.9155	-1.8033	0.0718
s3b2	-0.00041	0.9129	-0.6004	0.5484
s3b3	-0.00109	0.9036	-1.6470	0.1000
s3b4	-0.00053	0.8991	-0.7732	0.4396
s3b5	-0.00062	0.8757	-0.7039	0.4817
s4b1	0.00012	0.9047	0.1587	0.8739
s4b2	-0.00154	0.9086	-2.3141	0.0209
s4b3	-0.00136	0.8975	-1.9848	0.0475
s4b4	-0.00024	0.8783	-0.3264	0.7441
s4b5	-0.00149	0.8551	-1.5876	0.1128
s5b1	-0.00007	0.9440	-0.1459	0.8840
s5b2	-0.00083	0.8866	-1.3037	0.1928
s5b3	-0.00048	0.8571	-0.6882	0.4915
s5b4	-0.00260	0.8766	-3.6008	0.0003
s5b5	-0.00040	0.7567	-0.3388	0.7348
Average	-0.0009	0.8896	-1.0954	0.3707

Across the 25 test portfolios, the alphas remain generally small, with most values being concentrated in a narrow range around zero. The average p-value rose from 3.057 to 3.707, yet several alphas remain statistically significant at 5% level. Portfolio s1b1 continues to exhibit the largest deviation from zero, with an alpha of -0.00507, although its p-value is now insignificant. Similar patterns to the pre-adjustment analysis are ob-

served for portfolios s2b1, s4b2, s4b3 and s4b5. These findings suggest that, despite the quality adjustment, the model's ability to price the test portfolios has not improved. The adjustment increases the relevance of the size factor itself, but it does not enhance the overall model performance, as the average alpha remains largely unchanged while the average adjusted R^2 decreases. Figure 12 visualizes these regression results, providing a clear graphical representation of the model's performance after the quality adjustment. Comparing this with figure 10 gives a clear view of the effect of the quality adjustment on the model's performance.

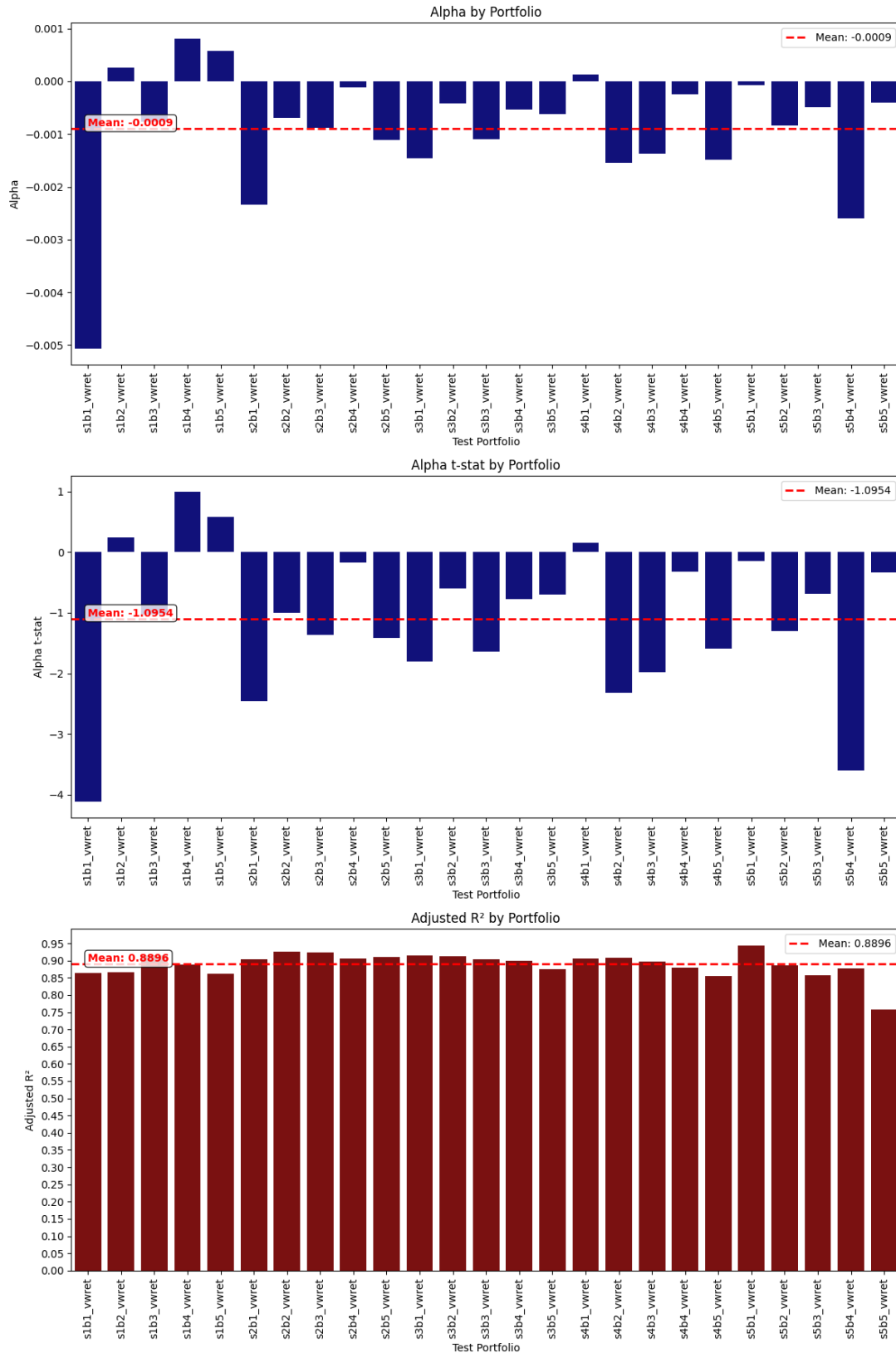


Figure 12: Reports alphas, alpha's t-statistics and adjusted R^2 of the regression used to test the post-adjustment model against the 25 5x5 test portfolios

In summary, while the introduction of the quality adjustment improves the characteristics of the SMB factor, raising its average and reducing its volatility, this effect does not translate into improved performance by the model as a whole.

5 Conclusion

This thesis set out to achieve three main objectives: first, to reproduce the Fama-French five-factor model over the period January 1972-December 2024; second, to evaluate the model's explanatory power on standard test portfolios; and third, to examine whether a quality adjustment to the SMB factor could improve both the factor itself and the overall performance of the model. The reproduction of the Fama-French five-factor model was successful. All five factors constructed in this study closely match their counterparts from Ken French's data library, as evidenced by high correlations, similar distributions and the Kolmogorov-Smirnov test results. This confirms that the methodology used for factor construction is accurate and that further analysis can reliably compare the original and adjusted versions of the model. Testing the original model on the 25 5x5 size-book-to-market test portfolios demonstrated that the Fama-French five-factor model continues to explain a large portion of the cross-section of stock returns, with an average adjusted R^2 of approximately 90%. While most alphas are small and not statistically significant, some portfolios, particularly those at the extremes of the size and book-to-market dimensions, exhibit significant residual returns. The SMB factor, in particular, shows a low average contribution, high volatility, and a pronounced January effect, confirming some of the concerns raised in prior literature regarding its robustness and economic relevance. Introducing the quality adjustment to the SMB factor improves its characteristics. Post-adjustment, the size factor shows a higher mean, lower volatility and a substantially reduced January effect. This suggests

that incorporating firm quality can meaningfully refine the size premium. However, the adjustment did not translate into improved overall model performance. The average adjusted R^2 decreases to 88.9% and the magnitude and statistical significance of the alphas remain largely unchanged across the test portfolios. This indicates that, while the quality adjustment enhances the SMB factor itself, the overall explanatory power of the Fama-French five-factor model remains untouched.

These findings have several implications. They demonstrate that while factor-specific adjustments, such as quality control for size, can improve the characteristics of individual factors, they do not necessarily lead to better model performance. Overall, this thesis contributes to the ongoing discussion on factor relevance in asset pricing by confirming the robustness of the Fama-French five-factor model, demonstrating the potential for quality adjustments to refine the size factor, and providing empirical evidence on the behavior of the size premium.

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